MATH6006: Statistical Methods for OR Modelling

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Introduction

In this lecture, we consider the following.

- Developing methods to derive inference on population parameters, e.g., mean and variance, from a sample.
- Understand the uncertainty in those estimates, e.g. find a range (interval) that contains the true value of each parameter with some level of confidence.

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- Developing methods to derive inference on population parameters, e.g., mean and variance, from a sample.
- Understand the uncertainty in those estimates, e.g. find a range (interval) that contains the true value of each parameter with some level of confidence.

Point estimation: estimate the values of population parameters using a sample.

Confidence interval estimation: estimate the interval that contains the true value of a population parameter.

Point Estimation

Suppose $X_1, \ldots, X_n$ are i.i.d. of a random variable X, which has mean μ and variance σ^2.
The sample mean

$$\overline{X} = \frac{1}{n}(X_1 + X_2 + \cdots + X_n)$$

is also a random variable and its mean is μ.

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Where X_i takes an actual value x_i, then the actual average

$$\overline{x} = \frac{1}{n}(x_1 + x_2 + \cdots + x_n)$$

is called the **estimate**.
The **estimate** is a particular value of the **estimator**, which is a random variable.

Unbiased Estimator

A good estimator will be unbiased.

Unbiased Estimator

A point estimator $\hat{\theta}$ is said to be an *unbiased estimator* of a population parameter θ if its expected value is equal to that parameter; that is, if

$$E[\hat{\theta}] = \theta,$$

then $\hat{\theta}$ is an unbiased estimator of θ.

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- The sample mean is an unbiased estimator of μ:

$$E[\overline{X}] = \mu.$$

- The sample variance is an unbiased estimator of σ^2:

$$E[S^2] = \sigma^2.$$

Bias of an Estimator

Definition: Bias

Let $\hat{\theta}$ be an estimator of θ. The **bias** in $\hat{\theta}$ is defined as the difference between its mean and θ,

$$\texttt{bias}(\hat{\theta}) = E[\hat{\theta}] - \theta.$$

The bias of an unbiased estimator is 0.

Consistency

An estimator $\hat{\theta}$ of a parameter θ is a function of a given sample $\{X_i\}$, and is a random variable.
We can write $\hat{\theta} = \hat{\theta}(X_1, ..., X_n)$ to indicate that $\hat{\theta}$ depends on $(X_1, ..., X_n)$.

Definition: Consistency

An estimator $\hat{\theta}$ is **consistent** if the expected squared error tends to zero as the size of the sample n tends to ∞, i.e.

$$E[(\hat{\theta} - \theta)^2] \to 0 \text{ as } n \to \infty.$$

Choosing an Estimator

Suppose we have two unbiased estimators $\hat{\theta}_1$ and $\hat{\theta}_2$ of an unknown parameter θ. We use the estimator with the smaller variance. For example, if

$$Var(\hat{\theta}_1) \leq Var(\hat{\theta}_2),$$

then $\hat{\theta}_1$ is preferred over $\hat{\theta}_2$.

Method of Moments

The method of moments can be useful in parameterising continuous distributions. An alternative, which is widely used, but we do not discuss here, is to maximise the likelihood.

The method of moments equates the sample moments (e.g. mean and variance) with the theoretical moments of the distribution being fitted.

Example

A sample $X_1, ..., X_n \sim exp(\lambda)$. We would like to estimate λ using the sample.
First, we compute the sample mean

$$\bar{x} = \frac{1}{n}(x_1 + x_2 + \ldots + x_n).$$

We then use the fact that $\mu = 1/\lambda$ to get

$$\bar{x} = \frac{1}{\hat{\lambda}}.$$

Hence, the estimated $\hat{\lambda}$ is

$$\hat{\lambda} = \frac{1}{\bar{x}}.$$

Fitting a Model with Two Parameters

If a model has two parameters, θ_1 and θ_2, then we we can say that

$$\mu(\hat{\theta}_1, \hat{\theta}_2) = \bar{X} \qquad \text{and} \qquad \sigma^2(\hat{\theta}_1, \hat{\theta}_2) = S^2.$$

We then solve the system of equations for $\hat{\theta}_1$ and $\hat{\theta}_2$.

For the normal distribution, this is straightforward as we can set $\mu = \bar{X}, \sigma^2 = S^2$.

Exercise: Estimate θ using the Method of Moments

Exercise

Let $x_1, x_2, \ldots, x_n$ be a random sample from a distribution with pdf

$$f(x;\theta) = \frac{1}{\theta} x^{(1-\theta)/\theta}, \qquad 0 < x < 1,$$

where $\theta \in \Theta = \{\theta | 0 < \theta < \infty\}$. Use the method of moments to estimate θ.

We first find the expected value using the distribution $f(x)$

$$\bar{x} = \int_0^1 x \frac{x^{(1-\theta)/\theta}}{\theta} dx$$

Organising the powers of x,

$$\bar{x} = \frac{1}{\theta} \int_0^1 x^{1/\theta} dx = \frac{1}{1+\theta}.$$

We can then rearrange to find the estimator $\hat{\theta} = \frac{1}{\bar{x}} - 1$.

Confidence Interval Estimation

A confidence interval (CI) is a method of expressing the accuracy of an estimate. Suppose $\hat{\theta} = \hat{\theta}(X_1, ..., X_n)$ is an estimator for θ, then if we can find functions U and L such that

$$P(\theta > U(\hat{\theta}) = U(X_1, ..., X_n)) = \alpha/2$$
$$P(\theta < L(\hat{\theta}) = L(X_1, ..., X_n)) = \alpha/2,$$

we say that $(L(x_1, ..., x_n), U(x_1, ..., x_n))$ is a 100(1 - α)% Confidence Interval (CI) for θ.
The most common form of CI is a 95% CI for the mean of a distribution.
(Note that the CI does not need to be symmetric.)

Confidence Interval for μ

1. The case when σ is known.

Where σ is known, the Central Limit Theorem allows us to construct a confidence interval for μ.

Let $X_1, ..., X_n$ be i.i.d. variable with mean μ and variance σ^2. We know that the r.v. $\overline{X}$ is an estimator for μ.

Standardize $\overline{X}$ and get $Z = \sqrt{n}(\overline{X} - \mu)/\sigma$.
Determine $z_{\alpha/2}$ such that

$$P(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}) = 1 - \alpha$$

or after rearranging,

$$P\left(\overline{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \leq \mu \leq \overline{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

Confidence Interval for μ

By the Central Limit Theorem for large n we have

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} \approx N(0, 1)$$

We can then use tables or a look up function to calculate $z_{\alpha/2}$ for a given α.

For example, for 95% confidence intervals and known σ, the confidence intervals can be written as

$$\left[\bar{x} - 1.96\frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96\frac{\sigma}{\sqrt{n}}\right]$$

Summary

1. The $100(1-\alpha)\%$ Confidence Interval (CI) is given by

$$\left[\bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}},\ \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right]$$

2. The **Margin of Error** (ME) (also known as the sample error) is

$$\texttt{ME} = z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

   Note that $\frac{\sigma}{\sqrt{n}}$ is called the **sample standard error**. The **Width** w is equivalent to twice the margin of error: $w = 2(\texttt{ME})$.

3. The **Upper Confidence Limit** (UCL) is

$$\texttt{UCL} = \bar{x} + \texttt{ME}$$

   and the **Lower Confidence Limit** (LCL) is

$$\texttt{LCL} = \bar{x} - \texttt{ME}.$$

Confidence Interval for μ

Taking $\alpha = 0.05$ we construct the 95% interval for μ :

$$\begin{aligned}
0.95 &= P(-1.96 < Z < 1.96) \\
&= P\left(-1.96 < \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} < 1.96\right) \\
&= P\left(-1.96\frac{\sigma}{\sqrt{n}} < \overline{X} - \mu < 1.96\frac{\sigma}{\sqrt{n}}\right) \\
&= P\left(\overline{X} - 1.96\frac{\sigma}{\sqrt{n}} < \mu < \overline{X} + 1.96\frac{\sigma}{\sqrt{n}}\right).
\end{aligned}$$

So $[\bar{x} - 1.96\sigma/\sqrt{n},\ \bar{x} + 1.96\sigma/\sqrt{n}]$ is a 95% CI for μ.

Example: Student Waiting Times

Example

Suppose that the waiting times for students at a university student service centre are normally distributed with known population standard deviation of 18 minutes. A random sample of 81 students in the center has a mean waiting time of 25 minutes. Find the sample standard error, the margin of error, and the upper and lower confidence limits, of a 95% CI for the population mean μ

Example: Student Waiting Times

We have the following information:

$$\bar{x} = 25, \qquad \sigma = 18, \qquad n = 81, \qquad \text{and} \qquad \alpha = 5\%.$$

- The sample standard error of the estimator $\overline{X}$ is

$$\frac{\sigma}{\sqrt{n}} = \frac{18}{\sqrt{81}} = 2.$$

- The margin of error is

$$\texttt{ME} = z_{\alpha/2}\frac{\sigma}{\sqrt{n}} = 1.96 \times 2 = 3.92.$$

- The upper confidence limit is

$$\texttt{UCL} = \bar{x} + \texttt{ME} = 25 + 3.92 = 28.92.$$

- The Lower confidence limit is

$$\texttt{LCL} = \bar{x} - \texttt{ME} = 25 - 3.92 = 21.08.$$

Interpreting the Confidence Intervals

The interpretation of the 95% CI is as follows.
This particular sampling of 81 students gives CI $[21.08,\ 28.92]$ but if we drew another sample of 81 students, the 95% CI would likely be different.

Suppose we take a large number of independent samples of 81 students from a very large population and we calculate for each sample the corresponding 95% CI. Then 95% of those intervals would contain the true value of the population mean.

In summary, the interpretation is that **in repeated sampling**, 95% of the time this interval will contain the true value of μ; the width of the CI about $\bar{x}$ gives us an idea of how reliable an estimate is.

Confidence Interval for μ

2. The case when σ is unknown

If the variance is unknown we use the random sample variance

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2.$$

The sample standard deviation S has uncertainty of its own, which means we cannot use the standard normal distribution to determine the confidence interval. Instead, we use a *student-t distribution* such that

$$\left[\bar{x} - t_{n-1,\alpha/2}\frac{s}{\sqrt{n}},\ \bar{x} + t_{n-1,\alpha/2}\frac{s}{\sqrt{n}}\right].$$

This works if the population is normally distributed.

Justification for the t-distribution

If $X_1, ..., X_n$ are i.i.d. and $N(\mu, \sigma^2)$, then $\overline{X}$ is normally distributed, i.e. $\overline{X} \sim N(\mu, \frac{\sigma^2}{n})$ exactly for all n. Let S^2 be the sample variance estimator. Then the random variable

$$T_n = \frac{\overline{X} - \mu}{\sqrt{\frac{S^2}{n}}} \sim t_{n-1}$$

follows a t-distribution with $n-1$ degrees of freedom. The t-distribution is similar to $N(0,1)$: it is symmetric and has mean 0. For a given $0 < \alpha < 1$, we define $t_{n-1,\alpha/2}$ (also known as the reliability factor) by

$$P\left(-t_{n-1,\alpha/2} \leq T_n \leq t_{n-1,\alpha/2}\right) = 1 - \alpha.$$

That is

$$F_{T_n}(-t_{n-1,\alpha/2}) = \frac{\alpha}{2} \qquad \text{or} \qquad t_{n-1,\alpha/2} = -F^{-1}_{T_n}(\alpha/2),$$

where $F_{T_n}(\cdot)$ is the cumulative distribution function of T_n.

Example: Confidence Intervals for a Normal Distribution

Example

Assume that the data $x_1, x_2, ..., x_{20}$ comes from a normal population with sample mean 0.6605 and sample standard deviation 0.0925. Obtain the 95% CI for the population mean μ.

We have the following information:

$$n = 20, \quad \bar{x} = 0.6605, \quad s = 0.0925.$$

Hence, the CI is

$$\begin{aligned} \text{CI} &= \left[0.6605 - t_{19,0.025}\frac{0.0925}{\sqrt{20}},\ 0.6605 + t_{19,0.025}\frac{0.0925}{\sqrt{20}}\right] \\ &= \left[0.6605 - 2.093\frac{0.0925}{\sqrt{20}},\ 0.6605 + 2.093\frac{0.0925}{\sqrt{20}}\right] \\ &= [0.6172,\ 07038], \end{aligned}$$

where the value $t_{19,0.025} = 2.093$ comes from the statistical tables.

Confidence Interval for a Proportion p

The results presented so far are for continuous random variables but they have counterparts for discrete random variables. For example, we can estimate the parameter p in the binomial distribution, remembering that a binomial variable is the sum of n independent Bernoulli random variables.

Definition: Sample Proportion

Let X be the number of successes in a binomial sample of n observations with the parameter p. The sample proportion is

$$\widehat{P} = \frac{X}{n}.$$

Confidence Interval for a Proportion p

As X is the sum of a set of n independent Bernoulli random variables, each with probability of success p, $\widehat{P}$ is the mean of a set of independent random variables. We can say that $\widehat{P}$ is the **point estimator** of p and

$$\hat{p} = x/n$$

is the **estimate**.

The Central Limit Theorem can also be used to derive confidence intervals for p using

$$\begin{aligned} E[\widehat{P}] &= p \\ \sigma_{\widehat{P}} &= \sqrt{\frac{p(1-p)}{n}}. \end{aligned}$$

Confidence Intervals for a Proportion p

If the sample size n is large, the random variable

$$Z = \frac{\widehat{P} - p}{\sigma_{\widehat{P}}}$$

is approximately distributed as the standard normal. This approximation is good if

$$np(1-p) > 5.$$

For a given α there is a percentage point $z_{\alpha/2}$ such that

$$P(-z_{\alpha/2} < Z < z_{\alpha/2}) = 1 - \alpha$$

This allows us to say that the $100(1-\alpha)\%$ CI for the parameter p is given by

$$\left[\hat{p} - z_{\alpha/2}\sqrt{\frac{\hat{p}\,(1-\hat{p})}{n}}, \quad \hat{p} + z_{\alpha/2}\sqrt{\frac{\hat{p}\,(1-\hat{p})}{n}}\right].$$

Example: Television Sets

Example

In a random sample of $n = 500$ families owning television sets in a village in Hampshire, it is found that $x = 340$ families subscribed to Netflix. Find a 95% CI for the actual proportion of families in this village who subscribe to Netflix.

Example: Television Sets

Example

In a random sample of $n = 500$ families owning television sets in a village in Hampshire, it is found that $x = 340$ families subscribed to Netflix. Find a 95% CI for the actual proportion of families in this village who subscribe to Netflix.

From the problem, we know that

$$n = 500, \qquad x = 340, \qquad \text{and} \qquad \alpha = 0.05.$$

Example: Television Sets

We already calculated from the standard normal that $z_{\alpha/2} = 1.96$ and

$$\hat{p} = \frac{340}{500} = \frac{17}{25}, \qquad \sqrt{\frac{\hat{p}\,(1-\hat{p})}{n}} = 0.0209.$$

Hence, either end of the confidence interval can be calculated as

$$\hat{p} - z_{\alpha/2}\sqrt{\frac{\hat{p}\,(1-\hat{p})}{n}} = 0.6391$$

and

$$\hat{p} + z_{\alpha/2}\sqrt{\frac{\hat{p}\,(1-\hat{p})}{n}} = 0.7209.$$

The 95% CI is therefore

$$[0.6391,\ 0.7209].$$